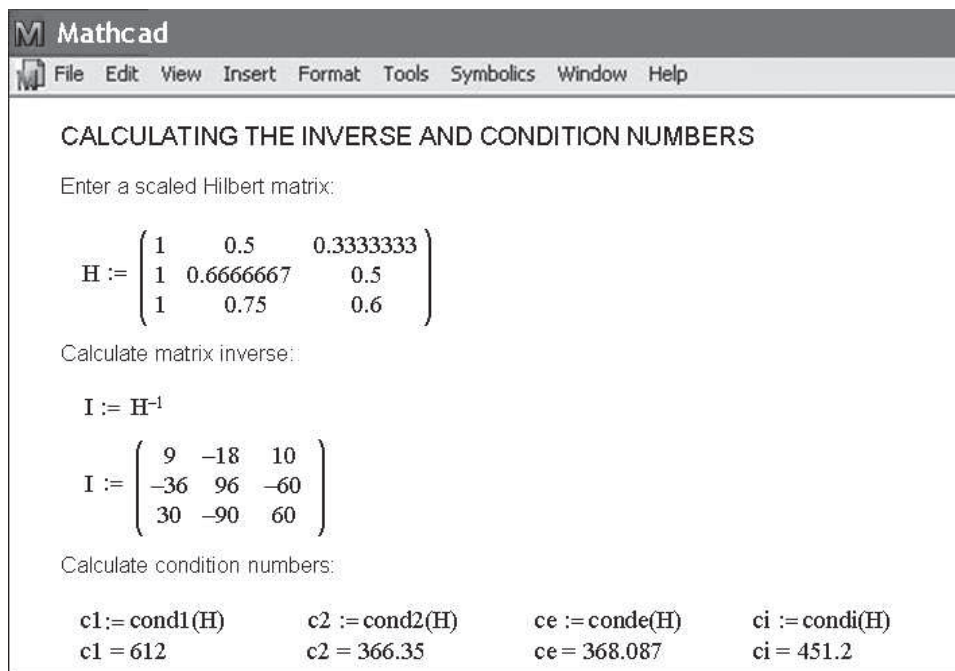


**FIGURE 11.9**

Mathcad screen to determine the matrix inverse and condition numbers of a scaled 3×3 Hilbert matrix.

PROBLEMS

11.1 Perform the same calculations as in (a) Example 11.1, and (b) Example 11.3, but for the tridiagonal system,

$$\begin{bmatrix} 0.8 & -0.4 & & \\ -0.4 & 0.8 & -0.4 & \\ & -0.4 & 0.8 & \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 41 \\ 25 \\ 105 \end{Bmatrix}$$

11.2 Determine the matrix inverse for Example 11.1 based on the LU decomposition and unit vectors.

11.3 The following tridiagonal system must be solved as part of a larger algorithm (Crank-Nicolson) for solving partial differential equations:

$$\begin{bmatrix} 2.01475 & -0.020875 & & \\ -0.020875 & 2.01475 & -0.020875 & \\ & -0.020875 & 2.01475 & -0.020875 \\ & & -0.020875 & 2.01475 \end{bmatrix} \begin{Bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{Bmatrix} = \begin{Bmatrix} 4.175 \\ 0 \\ 0 \\ 2.0875 \end{Bmatrix}$$

Use the Thomas algorithm to obtain a solution.

11.4 Confirm the validity of the Cholesky decomposition of Example 11.2 by substituting the results into Eq. (11.2) to see if the product of $[L]$ and $[L]^T$ yields $[A]$.

11.5 Perform the same calculations as in Example 11.2, but for the symmetric system,

$$\begin{bmatrix} 6 & 15 & 55 \\ 15 & 55 & 225 \\ 55 & 225 & 979 \end{bmatrix} \begin{Bmatrix} a_0 \\ a_1 \\ a_2 \end{Bmatrix} = \begin{Bmatrix} 152.6 \\ 585.6 \\ 2488.8 \end{Bmatrix}$$

In addition to solving for the Cholesky decomposition, employ it to solve for the a 's.

11.6 Perform a Cholesky decomposition of the following symmetric system by hand,

$$\begin{bmatrix} 8 & 20 & 15 \\ 20 & 80 & 50 \\ 15 & 50 & 60 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 250 \\ 100 \end{Bmatrix}$$

11.7 Compute the Cholesky decomposition of

$$[A] = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 25 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

Do your results make sense in terms of Eqs. (11.3) and (11.4)?

11.8 Use the Gauss-Seidel method to solve the tridiagonal system from Prob. 11.1 ($\varepsilon_s = 5\%$). Use overrelaxation with $\lambda = 1.2$.

11.9 Recall from Prob. 10.8, that the following system of equations is designed to determine concentrations (the c 's in g/m^3) in a series of coupled reactors as a function of amount of mass input to each reactor (the right-hand sides in g/d),

$$\begin{aligned} 15c_1 - 3c_2 - c_3 &= 3800 \\ -3c_1 + 18c_2 - 6c_3 &= 1200 \\ -4c_1 - c_2 + 12c_3 &= 2350 \end{aligned}$$

Solve this problem with the Gauss-Seidel method to $\varepsilon_s = 5\%$.

11.10 Repeat Prob. 11.9, but use Jacobi iteration.

11.11 Use the Gauss-Seidel method to solve the following system until the percent relative error falls below $\varepsilon_s = 5\%$,

$$\begin{aligned} 10x_1 + 2x_2 - x_3 &= 27 \\ -3x_1 - 6x_2 + 2x_3 &= -61.5 \\ x_1 + x_2 + 5x_3 &= -21.5 \end{aligned}$$

11.12 Use the Gauss-Seidel method (a) without relaxation and (b) with relaxation ($\lambda = 0.95$) to solve the following system to a tolerance of $\varepsilon_s = 5\%$. If necessary, rearrange the equations to achieve convergence.

$$\begin{aligned} -3x_1 + x_2 + 12x_3 &= 50 \\ 6x_1 - x_2 - x_3 &= 3 \\ 6x_1 + 9x_2 + x_3 &= 40 \end{aligned}$$

11.13 Use the Gauss-Seidel method (a) without relaxation and (b) with relaxation ($\lambda = 1.2$) to solve the following system to a tolerance of $\varepsilon_s = 5\%$. If necessary, rearrange the equations to achieve convergence.

$$\begin{aligned} 2x_1 - 6x_2 - x_3 &= -38 \\ -3x_1 - x_2 + 7x_3 &= -34 \\ -8x_1 + x_2 - 2x_3 &= -20 \end{aligned}$$

11.14 Redraw Fig. 11.5 for the case where the slopes of the equations are 1 and -1 . What is the result of applying Gauss-Seidel to such a system?

11.15 Of the following three sets of linear equations, identify the set(s) that you could not solve using an iterative method such as Gauss-Seidel. Show using any number of iterations that is necessary that your solution does not converge. Clearly state your convergence criteria (how you know it is not converging).

Set One	Set Two	Set Three
$8x + 3y + z = 12$	$x + y + 5z = 7$	$-x + 3y + 5z = 7$
$-6x + 7z = 1$	$x + 4y - z = 4$	$-2x + 4y - 5z = -3$
$2x + 4y - z = 5$	$3x + y - z = 4$	$2y - z = 1$

11.16 Use the software package of your choice to obtain a solution, calculate the inverse, and determine the condition number (without scaling) based on the row-sum norm for

(a)

$$\begin{bmatrix} 1 & 4 & 9 \\ 4 & 9 & 16 \\ 9 & 16 & 25 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 14 \\ 29 \\ 50 \end{Bmatrix}$$

(b)

$$\begin{bmatrix} 1 & 4 & 9 & 16 \\ 4 & 9 & 16 & 25 \\ 9 & 16 & 25 & 36 \\ 16 & 25 & 36 & 49 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 30 \\ 54 \\ 86 \\ 126 \end{Bmatrix}$$

In both cases, the answers for all the x 's should be 1.

11.17 Given the pair of nonlinear simultaneous equations:

$$\begin{aligned} f(x, y) &= 4 - y - 2x^2 \\ g(x, y) &= 8 - y^2 - 4x \end{aligned}$$

(a) Use the Excel Solver to determine the two pairs of values of x and y that satisfy these equations.

(b) Using a range of initial guesses ($x = -6$ to 6 and $y = -6$ to 6), determine which initial guesses yield each of the solutions.

11.18 An electronics company produces transistors, resistors, and computer chips. Each transistor requires four units of copper, one unit of zinc, and two units of glass. Each resistor requires three, three, and one units of the three materials, respectively, and each computer chip requires two, one, and three units of these materials, respectively. Putting this information into table form, we get:

Component	Copper	Zinc	Glass
Transistors	4	1	2
Resistors	3	3	1
Computer chips	2	1	3

Supplies of these materials vary from week to week, so the company needs to determine a different production run each week. For example, one week the total amounts of materials available are 960 units of copper, 510 units of zinc, and 610 units of glass. Set up the system of equations modeling the production run, and use Excel, MATLAB, or Mathcad, to solve for the number of transistors, resistors, and computer chips to be manufactured this week.

11.19 Use MATLAB or Mathcad software to determine the spectral condition number for a 10-dimensional Hilbert matrix. How many digits of precision are expected to be lost due to ill-conditioning? Determine the solution for this system for the case where each element of the right-hand-side vector $\{b\}$ consists of the summation of the coefficients in its row. In other words, solve for the case where all the unknowns should be exactly one. Compare the resulting errors with those expected based on the condition number.

11.20 Repeat Prob. 11.19, but for the case of a six-dimensional Vandermonde matrix (see Prob. 10.17) where $x_1 = 4$, $x_2 = 2$, $x_3 = 7$, $x_4 = 10$, $x_5 = 3$, and $x_6 = 5$.

11.21 Given a square matrix $[A]$, write a single line MATLAB command that will create a new matrix $[Aug]$ that consists of the original matrix $[A]$ augmented by an identity matrix $[I]$.

11.22 Write the following set of equations in matrix form:

$$50 = 5x_3 - 7x_2$$

$$4x_2 + 7x_3 + 30 = 0$$

$$x_1 - 7x_3 = 40 - 3x_2 + 5x_1$$

Use Excel, MATLAB, or Mathcad to solve for the unknowns. In addition, compute the transpose and the inverse of the coefficient matrix.

11.23 In Sec. 9.2.1, we determined the number of operations required for Gauss elimination without partial pivoting. Make a similar determination for the Thomas algorithm (Fig. 11.2). Develop a plot of operations versus n (from 2 to 20) for both techniques.

11.24 Develop a user-friendly program in either a high-level or macro language of your choice to obtain a solution for a tridiagonal system with the Thomas algorithm (Fig. 11.2). Test your program by duplicating the results of Example 11.1.

11.25 Develop a user-friendly program in either a high-level or macro language of your choice for Cholesky decomposition based on Fig. 11.3. Test your program by duplicating the results of Example 11.2.

11.26 Develop a user-friendly program in either a high-level or macro language of your choice for the Gauss-Seidel method based on Fig. 11.6. Test your program by duplicating the results of Example 11.3.

11.27 As described in Sec. PT3.1.2, linear algebraic equations can arise in the solution of differential equations. For example, the following differential equation results from a steady-state mass balance for a chemical in a one-dimensional canal,

$$0 = D \frac{d^2 c}{dx^2} - U \frac{dc}{dx} - kc$$

where c = concentration, t = time, x = distance, D = diffusion coefficient, U = fluid velocity, and k = a first-order decay rate. Convert this differential equation to an equivalent system of simultaneous algebraic equations. Given $D = 2$, $U = 1$, $k = 0.2$, $c(0) = 80$ and $c(10) = 20$, solve these equations from $x = 0$ to 10 with $\Delta x = 2$, and develop a plot of concentration versus distance.

11.28 A *pentadiagonal* system with a bandwidth of five can be expressed generally as

$$\begin{bmatrix} f_1 & g_1 & h_1 & & & \\ e_2 & f_2 & g_2 & h_2 & & \\ d_3 & e_3 & f_3 & g_3 & h_3 & \\ & & \cdot & \cdot & \cdot & \\ & & & \cdot & \cdot & \cdot \\ & & & & \cdot & \cdot & \cdot \\ & & & & & d_{n-1} & e_{n-1} & f_{n-1} & g_{n-1} \\ & & & & & & d_n & e_n & f_n \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \cdot \\ \cdot \\ \cdot \\ x_{n-1} \\ x_n \end{Bmatrix} = \begin{Bmatrix} r_1 \\ r_2 \\ r_3 \\ \cdot \\ \cdot \\ \cdot \\ r_{n-1} \\ r_n \end{Bmatrix}$$

Develop a program to efficiently solve such systems without pivoting in a similar fashion to the algorithm used for tridiagonal matrices in Sec. 11.1.1. Test it for the following case:

$$\begin{bmatrix} 8 & -2 & -1 & 0 & 0 \\ -2 & 9 & -4 & -1 & 0 \\ -1 & -3 & 7 & -1 & -2 \\ 0 & -4 & -2 & 12 & -5 \\ 0 & 0 & -7 & -3 & 15 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{Bmatrix} = \begin{Bmatrix} 5 \\ 2 \\ 0 \\ 1 \\ 5 \end{Bmatrix}$$